

PRODUCT · UFS FAMILY · IP67

UFS IP67

IP67-rated Direct Conversion Linear Detector Platform

The IP67-protected version of the Athlos UFS scanning array — same direct-conversion CdTe core, packaged in a fully sealed, dust-tight and water-resistant enclosure for harsh industrial X-ray inspection environments.



1 Key highlights

DETECTOR CdTe direct conversion	PLATFORM Linear scanning array	MODE TDI · Gen 3 Energy Integration
SPEED Up to 75 000 lps · 1700 m/min	INTEGRATION OEM-ready · plug-and-play	ENCLOSURE IP67 · dust-tight · water-resistant

2 Standard variants

UFS 150 IP67154.2 mm	UFS 225 IP67231.3 mm	UFS 460 IP67462.6 mm
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3 Technical specifications

Detector & pixel3.1				Active area & collimator3.2			
Detector material	CdTe	—	Bump-bonded to full-custom ASIC	Active width	2.8	mm	28 stages TDI
Pixel size	100 × 100	µm	Square pixel	Active length · UFS 150 IP67	154.2	mm	
Detector technology	Direct conversion	—	No intermediate scintillator	Active length · UFS 225 IP67	231.3	mm	
Pixel mode of operation	Gen 3 Energy Integration · TDI	—	Single-photon sensitive	Active length · UFS 460 IP67	462.6	mm	
Sensor mode	TDI	—	28 stages	Custom active length	up to 12	m	Built to order
Pixel binning (TDI)	1×1 / 2×2 / 4×4	pixel		Distance window → detector	7	mm	
Dynamic range	16	bit		Collimator · low energy	2 mm Cu	10 - 120 kV	Copper
Energy range	10 - 360	kV	LINAC up to 9 MeV on request	Collimator · high energy	2 mm W	120 - 360 kV	Tungsten

4 Application areas

Harsh / washdown environments - Food & beverage — wet, cold, washdown production lines - Pharmaceutical NDT — controlled-environment compliance - Outdoor / partially-enclosed industrial X-ray bays - Dusty mining, recycling, bulk-material inspection	Industrial inline inspection - Food safety & quality — foreign-object, seal, fill - Automotive — tires, castings, weld & pipe inspection - Electronics & semiconductor — PCB, reel, SMT, AXI - Cargo, train & container inspection
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Technical specifications (continued)

Scanning & readout3.3				Power, mechanical & environment3.4			
Object speed · 100 µm	up to 28	m/min	High-resolution tier	Sensor power supply	9 – 24	V DC	
Object speed · 200 µm	up to 180	m/min	Balanced tier	Power consumption	1.3	W/cm	Peak per cm of active length
Object speed · 400 µm	up to 1700	m/min	Ultra-fast tier	Operating temperature	10 – 40	°C	No Peltiers · no active cooling
Object speed	User-adjustable	–		Storage temperature	0 – 50	°C	
Line rate	1 – 75 000	lps		Physical dimensions (X·Y·Z)	255 · 160 · 50	mm	Without mounting brackets
Calibration	Factory + auto	–	Auto-adjusted within operating range	Weight	3.5	kg	UFS 225 IP67 reference
Communication interface	Gigabit Ethernet	–		Mounting brackets	Customisable	–	Flush with X-ray window on demand
Cable length	up to 10	m	Shielded · longer on request	Enclosure / IP rating	IP67	IEC 60529	Dust-tight · protected against temporary water immersion

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Software & integration

The **Athlos SDK** includes the C/C++ API for camera control, header files, libraries, documentation and example code. A graphical user interface is provided at no extra cost for plug-and-play evaluation of the sensor and on-board post-processing filters. Calibration profiles ship on-board; no flat-field correction or per-material setup is required.

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Ordering & configuration

Order codes6.1			
UFS 150 IP67	UFS-150IP67NC-44-TDI-HVS	154.2 mm	IP67 enclosure
UFS 225 IP67	UFS-225IP67NC-44-TDI-HVS	231.3 mm	IP67 enclosure
UFS 460 IP67	UFS-460IP67NC-44-TDI-HVS	462.6 mm	IP67 enclosure
Custom length	UFS-CUSTIP67NC-44-TDI-HVS	up to 12 m	Built to order · contact Athlos sales

DOCUMENT	REVISION	STATUS	ORIGIN
UFS-IP67-DS-2026.1	A · May 2026	Production release	Athlos Oy · Finland

NOTE

A **standard (non-IP67) version** of this detector platform is available as **UFS** for indoor, protected and laboratory environments — with identical imaging performance. Request the dedicated UFS datasheet for full specifications.

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Contact

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